L01 LEDs only work one way

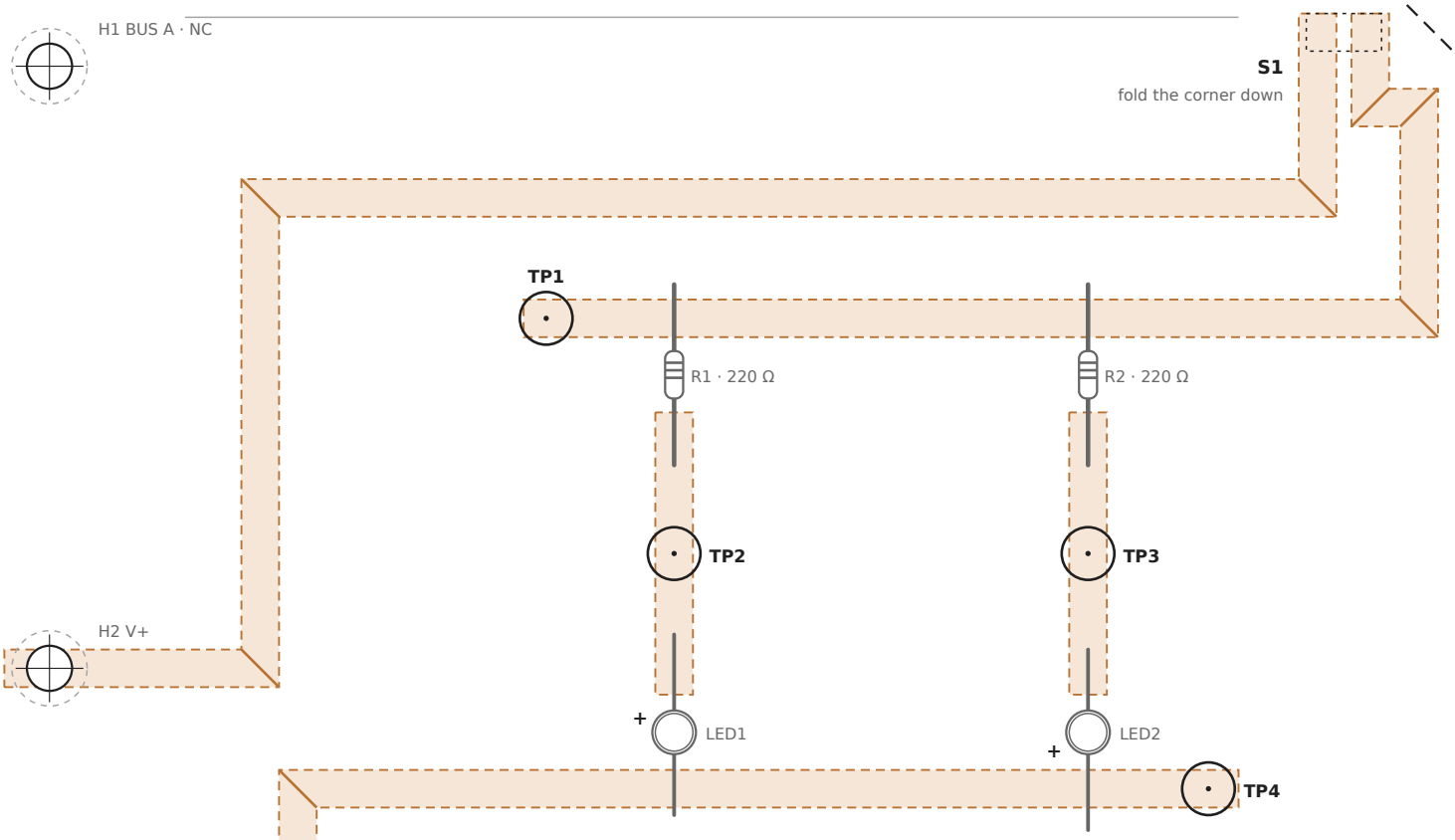
L01

Close the loop, then find out why an LED cares which way round it goes.

Draft: tagged steps may change

H1 BUS A · NC

S1
fold the corner down



H2 V+

H3 GND

You need

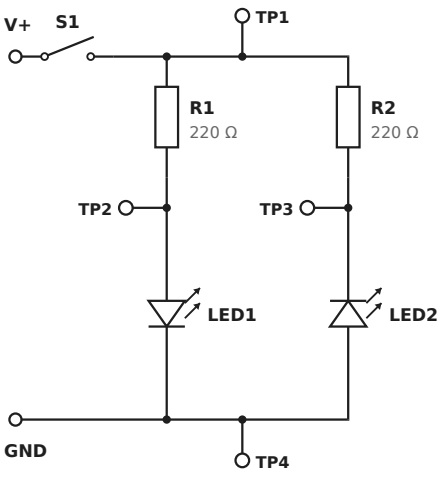
- 2 red LEDs, 5 mm
- 2 resistors, 220 Ω (red-red-brown) **[P]**
- 5 mm copper tape
- Power page P1 in the binder
- A multimeter for Measure

Build it

1. Lay the five lanes, one piece of tape each. Fold at the creases; never cut.
2. Stick a 10 mm piece of copper tape on the corner flap, S1. **[T0]**
3. Punch the holes and set eyelets in H2 and H3. **[T0]**
4. Place R1 and R2 across their gaps. Either way round works.
5. Place LED1 and LED2 **exactly as printed**: long leg on the +.
6. Hold each lead flat on its lane with a small patch of copper tape. **[T0]**
7. Bind this page with P1, then fold the corner down to switch on.

[T0] and **[P]** mark choices that may change: **[T0]** after the contact tests, **[P]** once the power supply is settled.

Schematic



V+ and GND come from P1 through the binder.

Predict LED2 is printed the other way round.

When you fold the corner down, which LEDs will light? Circle one for each.

LED1 lights / stays dark **LED2** lights / stays dark

Why do you think so?

Measure Hold the corner down. Black probe on TP4, red on each point.

TP1 _____ V **TP2** _____ V **TP3** _____ V

Lit? **LED1** Y / N **LED2** Y / N

Explain Why does only one LED light?

What would you change to make the other one light?

H4 BUS B · NC

Print at 100 %: this bar should measure 100 mm.